```
clear;
clc;
close all;

Fs = 5000;
T = 1/Fs;
fc = 1000;
Omega_c = 2*pi*fc;

% Analog Filter Prototype
% H(s) = Omega_c / (s + Omega_c)
b_a = [Omega_c];
a_a = [1 Omega_c];

% 1. Impulse Invariance Method
[b_imp, a_imp] = impinvar(b_a, a_a, Fs);

% 2. Bilinear Transformation with Prewarping
Omega_p = (2/T)*tan(Omega_c*T/2);

b_a_pre = [Omega_p];
a_a_pre = [1 Omega_p];

[b_blt, a_blt] = bilinear(b_a_pre, a_a_pre, Fs);

% Display Numerical Outputs
fprintf('=== EXPERIMENT 3: IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION
===\n\n');
```

```
=== EXPERIMENT 3: IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION ===
```

```
fprintf('Analog Cutoff Frequency (Omega_c): %.2f rad/s\n', Omega_c);
```

```
Analog Cutoff Frequency (Omega_c): 6283.19 rad/s
```

```
fprintf('Prewarped Analog Frequency (Omega_p): %.2f rad/s\n\n', Omega_p);
```

```
Prewarped Analog Frequency (Omega_p): 7265.43 rad/s
```

```
fprintf('--- IMPULSE INVARIANCE COEFFICIENTS ---\n');
```

```
--- IMPULSE INVARIANCE COEFFICIENTS ---
```

```
fprintf('Digital Numerator (b_imp):\n');
```

```
Digital Numerator (b_imp):
```

```
disp(b_imp);
```

```
    1.2566
```

```
fprintf('Digital Denominator (a_imp):\n');
```

```
Digital Denominator (a_imp):
```

```
disp(a_imp);
```

```
    1.0000   -0.2846
```

```
fprintf('--- BILINEAR TRANSFORMATION COEFFICIENTS ---\n');
```

```
--- BILINEAR TRANSFORMATION COEFFICIENTS ---
```

```
fprintf('Digital Numerator (b_blt):\n');
```

```
Digital Numerator (b_blt):
```

```
disp(b_blt);
```

```
    0.4208    0.4208
```

```
fprintf('Digital Denominator (a_blt):\n');
```

```
Digital Denominator (a_blt):
```

```
disp(a_blt);
```

```
    1.0000   -0.1584
```

```
% Frequency Response
[H_imp, f] = freqz(b_imp, a_imp, 1024, Fs);
[H_blt, ~] = freqz(b_blt, a_blt, 1024, Fs);

% Magnitude Response Comparison
figure('Name', 'Impulse Invariance vs Bilinear Transformation');

subplot(2,1,1);
plot(f, abs(H_imp), 'b', 'LineWidth', 1.5);
hold on;
plot(f, abs(H_blt), 'r--', 'LineWidth', 1.5);
grid on;
legend('Impulse Invariance', 'Bilinear Transformation (Prewarped)');
title('Magnitude Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Absolute Magnitude');

% Phase Response Comparison
subplot(2,1,2);
plot(f, angle(H_imp)*180/pi, 'b', 'LineWidth', 1.5);
hold on;
plot(f, angle(H_blt)*180/pi, 'r--', 'LineWidth', 1.5);
grid on;
legend('Impulse Invariance', 'Bilinear Transformation');
```

```
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```

